# HEALTH COMMODITIES QUANTIFICATION GUIDELINESMAINLAND TANZANIA

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## FOREWORD

The Government of Tanzania (GoT) through the Ministry of Health (MoH) and President's Office, Regional Administration and Local Government (PO-RALG) recognizes the importance of improving health services delivery to its people as a means to combat poverty. The GoT has developed policies and mapped out strategies to ensure that there is improvement of public health service delivery. The National Health Policy of Tanzania (2007) advocates for the provision of safe, quality, and effective medicines at affordable prices to all Tanzanians. To ensure an uninterrupted supply of essential health commodities, the Ministry of Health must accurately quantify the requirements needed to serve its citizens.

To address the need for a continuous supply of health commodities, the Ministry of Health conducted a holistic supply chain review in March 2017. The review found weaknesses in essential health commodities' quantification and recommended a bottom-up quantification approach whereby health facilities become the source of demand forecasts for essential health commodities.

The Ministry developed the Essential Health Commodities Quantification Guidelines in 2018 to assist key actors take a step-by-step approach in quantifying health commodity requirements and costs. The guideline has been used for demand forecasting of health commodities since 2019 to date. However, the Ministry of Health noted the need to review and update the quantification guideline to align with changes in the

supply chain landscape which included automation of the quantification process and improvements in the quantification steps. The need for review was also informed by observations from key recommendations of the 2022 BUQ assessment, field experiences, supportive supervision, and the 2023 BUQ Survey.

I therefore urge all stakeholders involved in the quantification exercise including Health facilities, Regional and Council Management Teams (RHMTs and CHMTs), and the National Quantification Team (NQT) to utilize these revised guidelines for effective quantification results.

**Prof. Tumaini J. Nagu**
**CHIEF MEDICAL OFFICER.**

## ACKNOWLEDGEMENT

Review of the Essential Health Commodities Quantification Guidelines has been a joint effort of stakeholders in the public health sector including implementing partners and other actors in the supply chain management system.

This revised guideline clearly outlines the levels involved in the health commodities quantification exercise, as well as the roles and responsibilities of each stakeholder in the reporting, monitoring, and evaluation process.

This work has been consultative, with the engagement of all the relevant parties in the health sector. On behalf of the Ministry, I would like to thank technical staff from the Ministry of Health (MoH); President's Office-Regional Administration and Local Governments (PO-RALG), Medical Stores Department (MSD). I also appreciate the support provided by Development and Implementing partners.

Last but not least, I would like to recognize the technical support from the Global Health Supply Chain Technical Assistance (GHSC-TA-TZ) project which was instrumental in accomplishing this important work.

**Daudi Msasi**
**CHIEF PHARMACIST**

## LIST OF ACRONYMS

| | |
|---|---|
| CCHP | Comprehensive Council Health Plan |
| CHMT | Council Health Management Team |
| eLMIS | Electronic Logistics Management Information System |
| EMT | Executive Management Team |
| FAMC | Forecasted average monthly consumption |
| GoT | Government of Tanzania |
| HMTC | Hospital Medicines and Therapeutic Committee |
| MIN-MAX | Minimum and Maximum inventory levels |
| MoF | Ministry of Finance |
| MoH | Ministry of Health |
| MSD | Medical Stores Department |
| MTC | Medicines and Therapeutic Committee |
| NDF | National Demand Forecast |
| NEMLIT | National Essential Medicines List |
| NQT | National Quantification Team |
| PMU | Procurement Management Unit |

| | |
|---|---|
| PORALG | President's Office of Regional Administration and Local Government Agency |
| R&R | Report and Request |
| RHMT | Regional Health Management Team |
| SDPs | Service Delivery Points |
| SMLEL | Standard Medical Laboratory Equipment List |
| SMLEG | Standard Medical Equipment Guidelines |
| SMRIEG | Standard Medical Radiology & Imaging Equipment Guidelines |
| STG | Standard Treatment Guideline |

# 1 BACKGROUND

The Ministry of Health (MoH), President's Office - Regional Administration and Local Government (PORALG), Medical Stores Department (MSD) in collaboration with Partners conducted a Holistic Health Supply Chain Review (2017) in which, under the area of quantification of health commodities, it was recommended to change the quantification approach from top-down to bottom-up in order to establish ownership and accountability for actors especially health facilities.

Bottom-up Quantification (BUQ) is designed to align with the necessity for health facilities to comply with the Public Procurement Act 2011 and its Regulations 2013 (GN 446), in conjunction with its amendments of 2022. This requirement mandates Health Facilities (HFs) to submit their annual demand forecasts to MSD by the 31st of January each year. This approach aims to establish a standardized system for accurately forecasting the needs of health commodities. This information is vital for informing government budgets, procurement decisions, and ensuring the uninterrupted provision of health services at service delivery points.

Since its inception in 2019, there have been robust improvements in the quantification processes which include full automation of the BUQ process that led to changes in the overall quantification process, roles and responsibilities of BUQ actors, techniques used in data preparation along with data flow for BUQ. There was also improved engagement of the zonal Medical Stores Department in Regional quantifications review meetings.

However, the Ministry of Health noted the need to review and update the quantification guideline to align with changes in the supply chain landscape which included: automation of the quantification process and improvements in the quantification steps. This was also informed by observations from key recommendations of the 2022 BUQ assessment, field experiences, supportive supervision, and the 2023 survey of the approach.

The 2023 BUQ guideline will facilitate standardization of the quantification process, quality control, improved transparency, and efficiency across all levels.

# 2 GOAL AND OBJECTIVES OF THE GUIDELINE

## 2.1 Goal of the guideline

The main goal is to guide healthcare workers to properly conduct the quantification process to enhance health commodities availability.

## 2.2 Objectives of the guideline

The main objective of this guideline is to guide key stakeholders of the health supply chain in the country to apply a step-by-step procedure in the annual quantification of health commodities.

The specific objectives for this guideline are:

1. To outline guidance on steps to be taken during the quantification of health commodities at each level.

2. To describe quantification governing structures and their respective roles and responsibilities

3. To guide users on how to conduct monitoring and evaluation of the key outputs of bottom-up quantification

4. To guide users on risk management of bottom-up quantification exercise

5. To establish a supply plan that will inform MSD procurement plan

The users of this guideline are Health Care Providers, National Quantification Team (NQT), Regional and Council Health Management Teams, Implementing Partners, Donor Agencies, Technical Advisors, and other key supply chain actors.

## 2.3 Definition of Terms

- **Quantification** is the process of estimating health commodity quantities and costs required for a specific health program and determining when the health commodities should be delivered to ensure an uninterrupted supply.

- **Health Commodities:** include pharmaceuticals, diagnostics, medical supplies, dental, and other products that may be needed for the provision of health services.

- **Bottom-up quantification** is an approach whereby the health facilities are the source of data and conduct the main role of projecting their demand forecasts of health commodities.

- **Demand Forecasting:** implies estimating the quantities of health commodities and cost required to meet clients' demands that will be used during a particular time frame.

- **Supply planning:** is a process that involves analyzing demand forecasts, comparing the forecasted demand to existing inventory, and coordinating the supply chain activities needed to meet that demand.

- **Program-Specific Health commodities:** Refer to medicines, laboratory supplies/reagents, hospital supplies, and other medical supplies that are procured, stored, and distributed for free to clients/patients for prevention, diagnosis, and/or treatment of a specific disease or group of diseases coordinated by the Vertical Program (s).

- **Vertical Programs:** initiatives, which focus on a particular disease or group of diseases.

- **Cross-cutting Essential Health Commodities:** includes essential medicines, laboratory supplies/ reagents, hospital supplies and other medical supplies

which can both be sold by MSD to SDPs but also provided for free by Vertical programs through MSD for prevention, diagnosis and/or treatment of a certain condition.

# 3 QUANTIFICATION GOVERNING STRUCTURES, ROLES, AND RESPONSIBILITIES

## 3.1 Quantification Governing Structures

The Quantification process will be conducted, guided and monitored by various structures within the healthcare delivery system. The actors in the BUQ process have been assigned clear roles and responsibilities to ensure smooth operationalization of the demand forecasting and supply planning activities. The Quantification governing structures exploit the existing structures within the MoH, PORALG and Ministry of Finance (MoF). The NQT will be formed and given the Terms of Reference (ToR) as stipulated in this guideline.

The structures responsible for management of the quantification process are:

1. Primary Health Facility (PHFs) Level
2. Council Level
3. Regional Level
4. President's Office - Regional Administrative and Local Government (PORALG)
5. Secondary & Tertiary Hospital Level

6. National Quantification Team (NQT)
7. Medical Stores Department (MSD)
8. Ministry of Health (MoH)
9. Ministry of Finance (MoF)

## 3.2 Roles and Responsibilities

### 3.2.1 Primary Health Facility Level

Primary Health Facilities (PHFs) are critical to the success of the National Demand Forecasts (NDF). Forecasts shall be owned at the facility level to ensure facilities have the commodities they require to serve their clients adequately.

**Roles and Responsibilities of Primary Health Facilities**

1. **Health Facility Medicines and Therapeutic Committee**
   i. Preparation of BUQ data.
   ii. Create the demand forecast report.
   iii. Verification of the annually adjusted consumptions.
   iv. Review demand forecast data populated by the system.
   v. Filling all sources of funds.
   vi. Produce written reports on the forecasting processes

vii. Submission of demand forecasts and report to the health facility in charge for authorization.

viii. Monitor procurement of health commodities based on the forecasted report

**2. Health Facility In-Charge**

i. Review and authorize the submitted demand forecasts

ii. Submit demand forecasts to the Council level by 31st August every year.

iii. Solicit resources to support BUQ activities and procurement of forecasted commodities.

**Note:** For the National Demand Forecasts (NDF) to be successful, health facilities must submit accurate Report and Request (R&R) forms timely to the Council on a monthly or bi-monthly basis. The accurate reporting of health facility-level consumption is the foundation of a successful forecast.

### 3.2.2 Council Level

The council level will comprise of District Executive Director (DED) and CHMT and each will perform different activities as described below;

**Roles and Responsibilities at the Council Level:**

1. **Council Health Management Team (CHMT)**

   i. Review and approve submitted Health Facilities' Demand Forecasts and document all assumptions and modifications done.

   ii. Capacitate healthcare workers from PHFs and other Hospitals at council level on the agreed approach for the annual NDF exercise.

   iii. Ensure the predetermined scope of demand forecast at council level is maintained according to the number and type of health facilities within their councils, and programs for emergencies.

   iv. Guided by the National Quantification Team, CHMTs will advise PHFs and other Hospitals at Council level to adopt the best demand forecast methods

   v. Co-opt other members as per quantification exercise requirements.

   vi. Obtain and keep records of demand forecast reports from all Health Facilities in the Council.

   vii. Produce written reports on the forecasting processes and the final results for submission to the RHMT office by 21st September of every year.

   viii. Monitor procurement of health commodities based on the forecasted report.

2. **District Medical Officer (DMO)**

   i. Appoint and assign duties to the Demand Forecast team.

   ii. Appoint the District Pharmacist to serve as secretary to the team and instruct him/her to prepare schedule of activities for the exercise and submit for consideration and approval

   iii. Review and submit final Council Demand Forecast report to DED's office for approval

   iv. Ensure BUQ activities are included in the Comprehensive Council Health Plan (CCHP)

3. **District Executive Director (DED)**

   i. Instruct the District Medical Officer to initiate the Demand Forecast exercise

   ii. Approve and submit Council Demand Forecast report with covering letter to RAS

   iii. Solicit resources to support BUQ activities and procurement of forecasted commodities.

### 3.2.3 Regional Level

The Regional level is responsible for ensuring Council Demand Forecasts are submitted within a given timeframe.

**Roles and Responsibilities at Regional Level:**

1. **Regional Health Management Team (RHMT)**

   i. Review and approve submitted Council Demand Forecast and explicitly document all assumptions and modifications done.

   ii. Provide guidance and technical support to CHMTs on demand forecast processes, tools and methods as far as demand forecast guideline and SOPs stipulate.

   iii. Obtain and keep records of demand forecast reports from all councils in the region.

   iv. Perform any other role (technical and/or administrative) as delegated by the National Quantification Team.

   v. Co-opt other members as per quantification exercise requirement including members from Zonal MSD

   vi. Produce written reports on the forecasting exercise and the final regional demand forecast for submission to PO-RALG by 14th of October of every year.

   vii. Monitor procurement of health commodities based on the forecasted report.

**2. Regional Medical Officer (RMO)**

i. Appoint and assign duties to the Demand Forecast team.

ii. Appoint the Regional Pharmacist to serve as secretary to the team and instruct him/her to prepare schedule of activities for the exercise and submit for consideration and approval.

iii. Approval of Regional Demand Forecast after RHMT review and share with RAS.

**3. Regional Administrative Secretary (RAS)**

i. Assign the Regional Medical Officer to initiate the Demand Forecast exercise.

ii. Approve and submit Regional Demand Forecast report with covering letter to PORALG.

iii. Solicit resources to support BUQ activities and procurement of forecasted commodities.

### 3.2.4 President's Office Regional Administration and Local Governments (PORALG)

PO-RALG is responsible for ensuring regional consolidated demand forecasts are of good quality and submitted within a given timeframe.

**Roles and Responsibilities of the PO-RALG**

i. Review and approve Regional Demand Forecast requirements and submit to MoH by 15th November of every year.

ii. Provide guidance and technical support to RHMTs on demand forecast processes.

iii. Obtain and keep records of demand forecast reports from all regions in the country.

iv. Facilitate timely availability of funds for procurement of essential health commodities.

v. Review and approve budgets for quantification activities from councils as budgeted in their CCHP.

vi. Establish a Team for review of Regional Demand Forecast data before submission to MoH

vii. Participate in monitoring of the supply plan of all health commodities including Vertical Program Commodities in collaboration with MoH and MSD

viii. Resource solicitation for review of forecasts at PO-RALG level before submission to MoH

ix. Perform any other role (technical and/or administrative) as may be advised by the Ministry of Health.

### 3.2.5 Secondary and Tertiary Hospitals

The Hospital Medicines and Therapeutic Committee (HMTC) shall serve as a technical arm for the Secondary and Tertiary hospital's Quantification exercise. These committees will be responsible for performing all duties outlined in the Medicines and Therapeutic Committee (MTC) Guidelines 2020, including the quantification of health commodities.

**Roles and responsibilities at Secondary and Tertiary Hospitals:**

1. **Hospital Medicine and Therapeutic Committee**
    i. Preparation of BUQ data.
    ii. Create the demand forecast report.
    iii. Verification of the annually adjusted consumptions.
    iv. Review demand forecast data populated by the system.
    v. Filling all sources of funds.
    vi. Produce written reports on the forecasting exercise and the final Hospital demand forecast for submission to MoH by the 14th of October of every year.
    vii. Submission of demand forecast to the Medical Officer In charge or Hospital Director for authorization.

viii. Perform any other duty related to demand forecast as assigned by the Hospital Management Committee.

ix. Monitor procurement of health commodities based on the forecasted report

**2. Medical Officer In-Charge or Hospital Director**

i. Review submitted demand forecast.

ii. Approval of Hospital Demand Forecast after HMTC review and share with MoH.

iii. Appoint the Hospital I Pharmacist to serve as secretary to the team and instruct him/her to prepare a schedule of activities for the exercise and submit it for consideration and approval.

**Note:** For the National Demand Forecasts (NDF) to be successful, health facilities must submit accurate Report and Request (R&R) forms timely to MSD on a monthly or bi-monthly basis. The accurate reporting of health facility-level consumption is the foundation of a successful forecast.

### 3.2.6 Ministry of Health (MoH)

MoH through the Pharmaceutical Services Unit (PSU) is responsible for consolidating the Demand Forecast and funding information for Secondary and Tertiary hospitals in the country. The consolidated forecasts are shared with the

NQT for the creation of the combined National Demand Forecast.

**Roles and Responsibilities of the Ministry:**

i. Review and approve Secondary and Tertiary hospitals-based Demand Forecast requirements by the 15th of November of every year.

ii. Provide guidance and technical support to Secondary and Tertiary hospitals on demand forecast processes, tools, and methods as far as Demand Forecast guideline and SOPs stipulate.

iii. Obtain and keep records of demand forecast reports from all Secondary and Tertiary hospitals in the country.

iv. Ensure timely availability of funds for procurement of essential health commodities.

v. Shall review and approve budgets for quantification activities

vi. Shall monitor Supply Plan of all health commodities including Vertical Program Commodities in collaboration with PORALG and MSD

vii. The Permanent Secretary MoH submits the NDF from NQT to MSD for supply plan and procurement by 31st December every year.

viii. The Permanent Secretary MoH will appoint the members of the National Quantification Team

(NQT) to oversee and guide the quantification exercise as stipulated in the Terms of Reference

ix. Resource solicitation to ensure commodity availability in the country and supply chain system strengthening.

x. Perform any other role (technical and/or administrative) as advised by the National Quantification Team.

### 3.2.7 National Quantification Team (NQT)

The team shall be appointed by the Permanent Secretary to the Ministry responsible for Health and shall oversee the quantification process of health commodities at all levels in the country.

**The team shall comprise of:**

1. Director of Pharmaceutical Services from the Ministry responsible for Health (Chairperson)

2. Three heads of components from the Pharmaceutical Services Unit (PSU)

3. Three members from PO-RALG (Health commodity, Diagnostics and Radiology services coordinator in charge)

4. Two national quantification coordinators from MoH and PORALG

5. Two members from the Directorate of Curative Services (DCS)

6. Two members from the Diagnostics and Health Care Technical Services Unit (DHCTSU) of the Ministry of Health

7. Three representatives of different skills mix (Finance, Demand Planning, and Procurement) from the Medical Stores Department (MSD)

8. Supply Chain representatives from Vertical Programs

9. Representatives of Supply Chain Implementing Partners working at the National level

10. Representatives from Development Partners

11. One member of the Budgeting Office from the Ministry of Finance

12. A representative from the National Medicines and Therapeutic Committee (NMTC)

13. Representatives from Regional Pharmacists and Laboratory Technologists

14. Representatives from Council Pharmacists and Laboratory Technologists

15. Any other co-opted member(s) may be added at the discretion of the Director of Pharmaceutical Services

**NQT Secretariat**

There will be a secretariat comprising of four (4) technical staffs (3 from PSU and 1 from PORALG)

**Roles and responsibilities of NQT:**

i. Review aggregated forecasts from MoH and PORALG

ii. Project a three-year national demand forecast

iii. Submit the NDF to the permanent secretary -MoH for approval by 7th December every year

iv. Monitor Supply Plan of all health commodities including Vertical Program Commodities in collaboration with MoH, PORALG and MSD

v. Provide strategic advice on quantification of essential health commodities

vi. Facilitate the training of RHMTs, CHMTs and MTCs to enable them acquire necessary competencies for undertaking the demand forecasting exercise

vii. Prepare an action plan for the national quantification process

viii. Prepare, review and/or approve quantification tools and/or data collection forms/datasheets…

ix. Liaise with MSD on matters relating to supply planning and delivery schedules.

x. Conduct forecast accuracy analysis for health commodities.

xi. Conduct quantification implementation review to assess effectiveness of the process in terms of forecast accuracy, supply planning, and funding.

xii. Perform any other role as may be assigned by the Permanent Secretary to the Ministry responsible for Health.

**Roles and responsibilities of NQT Secretariat:**

a. Aggregate and consolidate all demand forecasts from MoH and PORALG to get the National annual demand forecast
b. Coordinating and organizing the NQT meetings by;
    i. Preparing agendas and meeting materials,
    ii. Taking minutes and soon after an NQT meeting
    iii. Managing communications between attendees
    iv. Ensuring that all logistical aspects of the meeting run smoothly.
    v. Follow-up on status of action plan, document status and share with NQT members
    vi. Organize and participate in NQT sub-committee meetings where needed
    vii. Perform any other role as may be assigned by the NQT.

### 3.2.8 Medical Stores Department (MSD)

MSD shall have the following roles and responsibilities in the quantification processes.

i. Receive the NDF from the permanent secretary - MoH for the preparation of the supply plan and the procurement plan.

ii. Liaise with the NQT on all matters relating to supply planning development.

iii. Adhere to the developed supply plan per product description, quantity and planned delivery schedule

iv. Review, update, and monitor the pipeline regularly.

v. Report to the NQT every quarter the status of the supply plan including potential risks that can affect commodity supply

vi. Liaise with MoH and PORALG on the availability of funds for procurements of health commodities.

vii. Participate and contribute as a National Quantification Team member in all scheduled meetings.

viii. Participate in review meetings with the respective regional teams

ix. Perform any other role as may be assigned by the NQT.

# 4 THE BOTTOM-UP QUANTIFICATION PROCESS

The quantification process shall have three main stages as below:

1. Preparation
2. Forecasting
3. Supply Planning

## 4.1 Preparation

For the purpose of this guideline, Bottom-Up Quantification preparation means activities that are done prior to the actual execution of the demand forecast process. This is an important phase that needs to be considered at all levels within the governance structure of the demand forecast process.

The bottom-up quantification process will begin on the 1st of August every year by the MoH informing all Bottom-up Quantification implementers to commence the process.

**Preparations at health facilities**

i. Convene the medicine therapeutic committee to oversee the quantification

ii. Review and verify aggregated data from the automated BUQ module in eLMIS

iii. Review the list of predetermined essential health commodities to be forecasted

iv. List the anticipated services, expansion or changes in service delivery expected in the coming financial year

v. Review the scope of clients whose products will be forecasted

vi. Collect data from accountants on previous commodity expenditures and projected funding that will be mobilized in the next financial year for commodity procurement

**Preparations at council level**

i. Convene council health management team to prepare for quantification

ii. Map out facilities that will be opened or closed and align with listed facilities in automated BUQ

iii. Inform health facilities on starting the bottom-up quantification process and provide oversight

**Preparations at MSD**

i. Convene respective experts including the demand planning and procurement team to prepare for supply planning

ii. Collect the most recent product prices to inform supply planning

iii. Update the master product list and align it with the supply planning tool

iv. Collect recent updates, and report on the status of incoming shipment

v. Collect procurement updates on the status of frameworks of contracts to inform lead time during supply planning

## 4.2 Demand Forecasting

This stage involves estimating the quantities of health commodities and cost required to meet client demand that will be used during a particular time frame. The demand forecasting stage will start by building the forecasting assumptions followed by a selection of the forecasting method and calculation of the forecasted consumption. This process will be done within a defined schedule for the forecast to be done and the forecast will be for one financial year.

### 4.2.1 Assumption Building

Assumptions play an important role in health commodities demand forecasts as it help to fill gaps in data and build projections that inform decision-making in the supply chain. This step involves building and consensus on the forecasting assumptions that will be selected. During assumption building, the following factors need to be considered:

- Historical data and trends: conduct a review of historical data and trends regarding various aspects including, consumption trends, disease patterns, resource flow, demographics, supply chain issues, and other factors to analyze and understand trends over

time with impact on demand and supply of health commodities

- External factors: consider external factors such as disease outbreaks, disasters, changes in policy and socio-economic development, and others that could impact the demand and supply of health commodities

- Stakeholders' consultation: engage with stakeholders to understand their perspective on any possible factors that may affect the demand and supply of health commodities.

To project forecasts, consider assumptions on future program performance, based on factors influencing demand for services and commodities

i. Population growth rate within a geographical area (increases needs for services)

ii. Demographic changes (e.g., higher rate of births the previous year than usual)

iii. Opening of a new facility at a nearby location (reduces the number of clients to be served)

iv. Expansion of the services offered at the health facility

v. Changes in national coverage (increase/ decreases demand for certain services)

vi. Outbreak trends (seasonal flu, communicable diseases)

vii. Changes in resources (specialists leaving the facility, specialists coming in, etc.)

viii. Adherence to treatment guidelines and testing algorithms

ix. Public Health marketing campaigns

### 4.2.2 Forecasting Methods

On deciding which forecasting method will be used, consider:

- Availability of adjusted consumption data considering the number of days stocked out
- Availability of morbidity data at the health facility
- Availability of issues data (proxy consumption data) that can be used
- Availability of service-level data

There are different methods for demand forecasting, which are classified according to the type of data used.

1. **Consumption Method**

   This is a quantification method that uses past consumption data to estimate future needs. This method is considered to be the most reliable, provided that accurate data has been used. It is best suited for stable programs or services. However, it can be challenging to use this method for new products or services.

The BUQ tool has been fully automated within the eLMIS system. It is equipped with the consumption data of each commodity for the last 12 months, taking into account the number of days the item was stocked out (adjusted consumption/usage). The tool requires health facility staff to review and verify the historical consumption/usage data from the eLMIS system with facility record-keeping data sources such as ILS Dispensing register, ARV Dispensing register, Rapid test kit register, Store ledger and Electronic Medical Records. For more information, please refer to the automated BUQ User Guide, along with the example provided in Annex 8.1.

**Steps to project the forecast using the consumption method.**

i. Indicate the forecasting assumption that has been selected for the respective product(s)

ii. Determine a quantity factor to assign to the forecasting assumption. Note the quantity factor is based on a consensus among the quantification team such as MTCs at the health facilities.

iii. The historical consumption data, per product, will be multiplied by the quantity factor (b above) to determine the projected forecast consumption of the next financial year.

## 2. Morbidity/Demographic Method

Population-based method, use of disease burden (number of cases), attendance, and incidence of common diseases

to estimate future needs to treat such cases. With the Morbidity method the number of clients or patients is first determined and thereafter Standard Treatment Guidelines (STG), Testing Algorithms or other conversion factors are available to convert several clients or patients to commodities required.

This is the most appropriate method for;

i. Products not previously stocked at the facility (no historical consumption data)

ii. New health commodities/services

iii. Rapidly changing programs or services

Projection of forecasts using the morbidity method will go through the conversion and projection stages as shown below:

**a. Conversion**

i. List the diseases or health conditions that were identified during the preparation stage and the number of cases recorded per disease or health condition

ii. Indicate all the products that will be administered in addressing the identified disease or health condition

iii. Determine the quantity of products that are expected to be administered or consumed for the disease or health condition, as indicated in the STG, SMLEG, SMLEL, SMRIEG, disease-specific

treatment guidelines or testing algorithms.

iv. Multiply the products (in ii above) per listed diseases or health conditions (in i above), by the quantity of products to be administered per guideline (in iii above). The result is the historical consumption that has been converted from number of cases of diseases or health conditions

**b. Projection**

i. Indicate the forecasting assumption that has been selected for the disease or health condition

ii. Determine a quantity factor to assign to the forecasting assumption.

iii. The historical consumption, per product, will be multiplied by the quantity factor (ii above) to determine the projected forecast consumption of the next financial year.

Data sources for validation: Reported cases in Electronic Medical Records such as; DHIS2, CTC 3 Macro database, Electronic TB & Leprosy Register (ETL).

### 3. Proxy Consumption Method

Use of issues data at the lowest level of supply chain (such as at health facility level) as proxy consumption and estimating the future needs based on such data. This method is used when actual consumption data is unavailable.

**4. Service Level Projection of Requirements**

Uses consumption data of health commodities from one setting to estimate the need in a similar setting. For example, using the average commodity needs for a dispensary in area A to estimate the needs for a dispensary in area B. This approach will consider the health facility in the second setting has similar characteristics in terms of type of facility, disease burden, catchment area, and types of services provided.

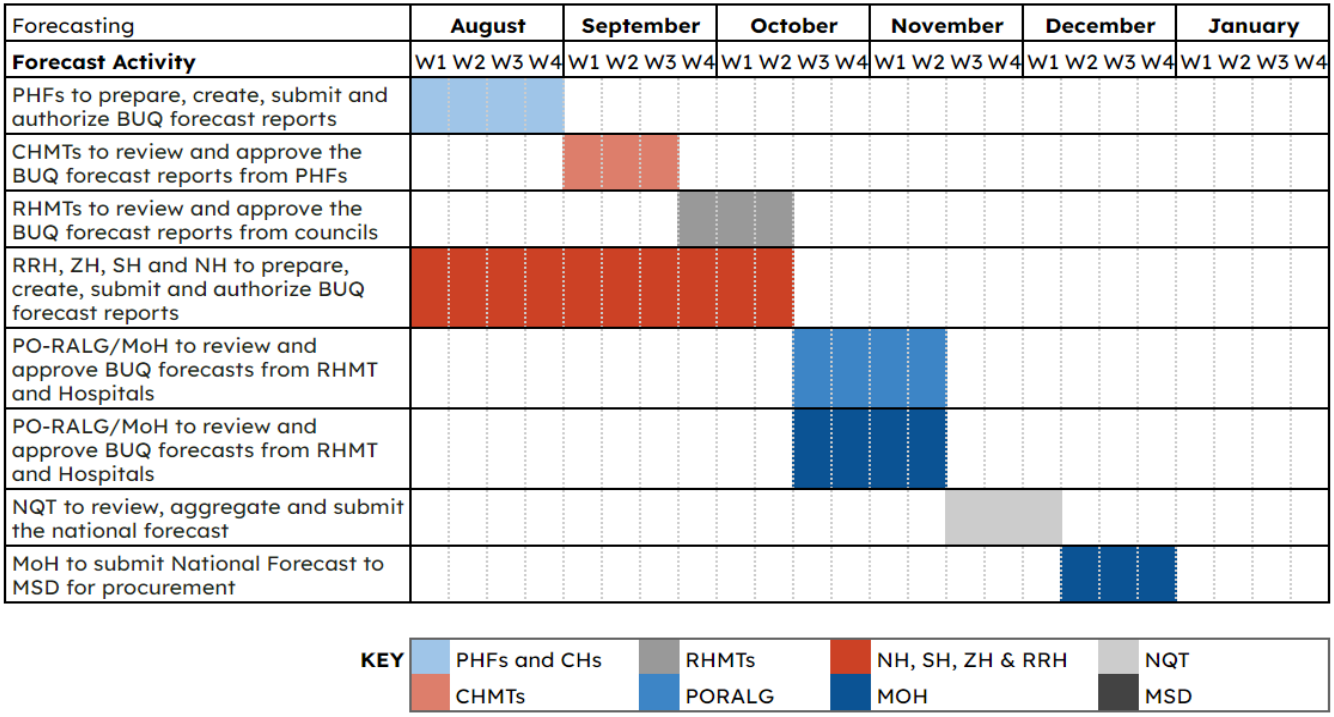
### 4.2.3 Forecasting for new facilities

This will be done by respective commodity managers with information on the type of facilities that are expected to be opened in the upcoming financial year of the BUQ forecast period. The service-level projection method can be used by considering other similar facility types. The recommended methods for forecasting shall be either consumption-based or morbidity/demographic-based methods depending on the nature of the health commodity/intervention and availability of data. However, in the event that consumption and morbidity/demographic data are not available, other methods as described above may be used. The demand forecast should not be adjusted according to the budget ceiling at the health facility level, this will allow resource mobilization and allocation to cover the health needs of the population. In the event that adjustment is required, the NQT will have to adjust according to the available information and circumstances.

### 4.2.4 Forecasting Schedule

The National Demand Forecast is an annual activity that will commence in August of every year and is led by the Chief Pharmacist. The forecasting exercise takes approximately five months from the initial stage at the health facility level to the approval of the nationally combined forecast. Find a detailed schedule in Figure 1.

## Figure 1 Gantt chart of forecasting



### 4.2.5 Forecast Period

Service delivery points (SDPs) will forecast their demand yearly, as stipulated above. However, the NQT will project the requirements (quantity and cost) for an additional two (2) years. Thus, they will submit 3-year demand forecasts for planning and resource mobilization to the Permanent Secretary of MoH. The Permanent Secretary MoH will approve and submit national annual forecasts to MSD for supply plan development. Quantification reviews will be done annually by the NQT using demand forecasts from service delivery points, as stipulated above

## 4.3 Supply Planning

Supply planning process involves determining the product quantities required to fill the supply pipeline, scheduling shipments, and estimating costs. The supply planning process follows after the completion of the national demand forecasting exercise. The approved National Demand Forecast, with a statement of requirements, shall be submitted to MSD by the Ministry of Health in December of every financial year.

The role of supply planning activity shall be managed and conducted by the Medical Stores Department (MSD) in collaboration with MoH, PO-RALG, and other supply chain stakeholders. For vertical programs commodities, MSD must work in collaboration with vertical programs in the management of supply plans for vertical commodities.

MSD as a government procuring entity shall use information

from the supply plan to develop procurement plan and call-off-orders as stipulated in the Public Procurement Act considering descriptions, specifications, statement of requirements, available stock on hand, shipments on order, agreed inventory control parameters (Maximum and Minimum stock levels), safety stock and other factors.

It should serve as a comprehensive guide for implementing and executing procurement based on Public Procurement Act 2011 and its Regulations 2013 as amended in 2016 and revised in 2022.

### 4.3.1 Purpose of Supply Planning

The purpose of supply planning is to provide quantities of shipments to procure based on a delivery schedule, and the total costs that will inform resource requirements.

The supply plan should be able to serve the following purpose:

- Indicate the quantity of shipment/s that needs to be procured. This quantity should be linked to inventory management by ensuring it maintains stock levels within the set MIN-MAX

- Provide a delivery schedule showing the desired delivery date. The delivery schedule will also be aligned with the stipulated MIN-MAX level at MSD and health facilities.

- Stipulate the total commodity cost based on shipments which will inform the funding status (gap or surplus).

### 4.3.2 Requirements for Supply Planning

The following are the requirements to consider when developing a supply plan:

i. Approved Forecasted Average Monthly Consumption (FAMC)

ii. Stock on hand at MSD and health facilities

iii. Quantities of near-to-expire commodities at MSD

iv. Commodities on order (incoming shipments that have already been procured)

v. MIN-MAX levels at MSD, at health facilities, and combined at the national level

vi. Supplier lead times

vii. Products Prices

viii. Commodities with framework agreements with suppliers

### 4.3.3 Process of Supply Planning

The process of supply planning involves a series of steps aimed at aligning supply with demand to ensure the availability of products when and where they are needed. The following are steps to be followed during supply planning:

**Step 1: Organize and Analyze the Data**

During this stage, organize and analyze data for supply

planning by removing errors, duplicates, or missing values which can lead to incorrect analysis and decision making. The information used in this stage includes the requirements listed in 4.3.2 above.

**Step 2: Build the Supply Planning Assumptions**

Supply planning involves making several assumptions to develop an effective plan. These assumptions serve as foundational principles and guide the decision-making process throughout the supply planning. It is essential to make assumptions that are context-specific and aligned with the requirements of the planning process.

**Step 3: Estimate the Total Commodity Requirements and Costs**

The supply planning tool capable of calculating the shipment quantities required and scheduling their delivery by considering the min-max will be used. This tool shall integrate data from various sources, such as health facility and MSD stock on hand, consumption data, MSD price product, and quantity on order to generate insights and facilitate decision-making.

Estimation of total commodity requirements involves the calculation of additional commodities needed to cover the procurement period while taking into consideration the required buffer stock, suppliers' lead times, available stock on hand, quantities already on order and any products that might expire before they can be used. The supply plan and delivery schedule may be adjusted to accommodate any challenges in storage and distribution.

The final supply plan should encompass the key components and information necessary for decision-making including shipment quantities, total cost for each item and delivery schedules.

**Step 4: Review and Monitoring of Supply Plan**

The review and monitoring of the supply plan involves regular assessments and adjustments to ensure effective management of the supply chain. The Medical Stores Department in collaboration with NQT and vertical programs should conduct quarterly reviews to evaluate the supply plan and make necessary modifications based on changes in consumption trends, delivery schedules, stock availability, and other relevant factors.

To enhance transparency and accessibility, the supply plan should be made visible to supply chain actors through a digitized MSD supply plan tool and report. This allows stakeholders to stay informed and aligned with the latest updates and adjustments in the supply plan.

In the case of vertical program commodities, MSD should maintain frequent communication and updates with the respective vertical programs, including sharing visibility into the MSD supply planning report. This includes providing information on the status of ordered and planned shipments, near-to-expire health commodities, distributed commodities, and any other pertinent details related to the vertical program's specific supply needs.

### 4.3.4 Schedule of Supply Planning

The supply planning process is an annual activity that will commence on the 1st of January every year and is led by MSD. It is a one-month activity, however, the developed supply plan is continuously monitored and reviewed as detailed in table 3 below.

**Table 3: Schedule of Supply Planning**

| SN | DATE | ACTIVITY | RESPONSIBLE |
|---|---|---|---|
| 1. | By December 31st | Receiving Approved National Requirements from Ministry of Health | MSD |
| 2 | January 1st -10th | Reviewing, analyzing and Organizing Specifications of Submitted Requirements and Incorporating Requirements in Supply Plan Tool (automated supply plan tool) | MSD |
| 3. | January 11-14th | Arranging Supply Planning Meeting and Stakeholder Invitation | MSD |
| 4 | January 15th -24th | Supply Planning Meeting | MSD, NQT, MoH, PORALG, MoF and other supply chain actors |
| 5 | January 25th | Submission of Supply Plan Results to EMT at MSD for approval | MSD |
| 6 | January 30th | Submission to PMU for procurement proceedings | MSD |
| 7 | Quarterly | Monitoring of supply planning implementation and updating NQT through quarterly meetings | MSD |

## Figure 2 Gantt chart of supply planning

| Supply planning | August | | | | September | | | | October | | | | November | | | | December | | | | January | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Forecast Activity** | W1 | W2 | W3 | W4 | W1 | W2 | W3 | W4 | W1 | W2 | W3 | W4 | W1 | W2 | W3 | W4 | W1 | W2 | W3 | W4 | W1 | W2 | W3 | W4 |
| Receiving Approved National Requirements from Ministry of Health | | | | | | | | | | | | | | | | | | | | | | | | |
| Reviewing, analyzing and Organizing Specifications of Submitted Requirements and Incorporating Requirements in Supply Plan Tool (automated supply plan tool) | | | | | | | | | | | | | | | | | | | | | | | | |
| Arranging Supply Planning Meeting and Stakeholder Invitation | | | | | | | | | | | | | | | | | | | | | | | | |
| Supply Planning Meeting | | | | | | | | | | | | | | | | | | | | | | | | |
| Submission of Supply Plan Results to EMT at MSD for management approval | | | | | | | | | | | | | | | | | | | | | | | | |
| Submission to PMU for procurement proceedings | | | | | | | | | | | | | | | | | | | | | | | | |
| Monitoring of supply planning implementation and updating NQT through quarterly meetings | | | | | | | | | | | | | | | | | | | | | | | | |

**KEY**

| | | | |
|---|---|---|---|
| PHFs and CHs | RHMTs | NH, SH, ZH & RRH | NQT |
| CHMTs | PORALG | MOH | MSD |

# 5 RISK MANAGEMENT

Risk management involves a systematic approach to identifying, assessing, and mitigating risks that could impact the availability of health commodities. In the quantification exercise, the responsible personnel are supposed to understand the risks that are associated with the forecasting and find the best way to mitigate any uncertainty.

There are different approaches in managing risk but before going into details it is better to understand the five steps that are involved in risk management.

1. **Risk Identification**

This step involves a comprehensive analysis to identify potential risks that could affect the forecasting and supply planning process. Risks can include inaccurate projections, disease outbreaks, shifts in program targets and policy changes. Other risks may involve supply chain disruptions due to natural disasters, transportation issues, changes in regulations, funding constraints, and issues related to product quality and availability.

2. **Risk Assessment**

Once the risks are identified, they need to be assessed in terms of their likelihood of occurrence and potential impact. The assessment helps to prioritize risks based on their significance.

3. **Mitigation Strategies**

Mitigation strategies aim to minimize the impact of identified risks. These strategies can include the identification of other solutions including establishing alternative sourcing options to mitigate supply chain disruptions and improving forecasting accuracy through data analysis and advanced modeling techniques. Implementing robust quality control measures can also facilitate the mitigation of risks at different steps of quantification. The MoH, PORALG, MSD, and other supply chain stakeholders should allocate adequate resources to enable the mitigation of identified risks.

4. **Monitoring and Evaluation**

Ongoing monitoring and evaluation are crucial for effective risk management. Regular reviews of the supply plan and quantification processes help identify areas of improvement and make necessary adjustments. This includes tracking the effectiveness of mitigation strategies, assessing the emergence of new risks, and adapting the risk management approach/es as needed. Timely monitoring and evaluation enable the MoH, PORALG, MSD and other stakeholders to proactively respond to changes and maintain an agile supply chain.

5. **Collaboration and Communication**

Effective risk management requires continuous communication and collaboration among all stakeholders involved in the quantification and supply planning process. This includes engagement with suppliers, vertical programs, regulatory bodies, and other relevant parties.

By integrating risk management practices into the quantification process, the MoH, PORALG, MSD and other stakeholders can proactively anticipate and address potential risks. This approach will lead to reliable forecasts and supply plans with reduced disruptions, improved product availability, and ultimately the ability to meet the healthcare needs of the population.

# 6 REPORTING, PERFORMANCE MONITORING AND EVALUATION

## 6.1 Reporting

Depending on the level, summarization of the data and reports shall be assigned within a realistic time frame to meet the scope of the national quantification process for Health Commodities.

### 6.1.2 Demand Forecast data flow

i. Health facilities will submit the demand forecasts to CHMT

ii. CHMT will submit council demand forecasts from PHFs and Council Hospitals to RHMT, which includes aggregated funding levels by facility

iii. RHMT will submit the final region demand forecast to PO-RALG.

iv. Regional, Referral, Zonal, Specialized and National Hospitals will submit their final demand forecasts to MoH

v. PO-RALG will submit final consolidated demand forecast from RHMT with funding information (Health Commodities financing sources) to NQT secretariat

vi. MoH will submit the approved national annual demand forecast to MSD

vii. MSD in collaboration with MoH and other key supply chain stakeholders will develop the supply plan based on the submitted national annual demand forecast.

**Note:** At all levels, the demand forecast shall be reviewed by using the 10 criteria for reviewing the demand forecast (Refer to annex 8.5 below)

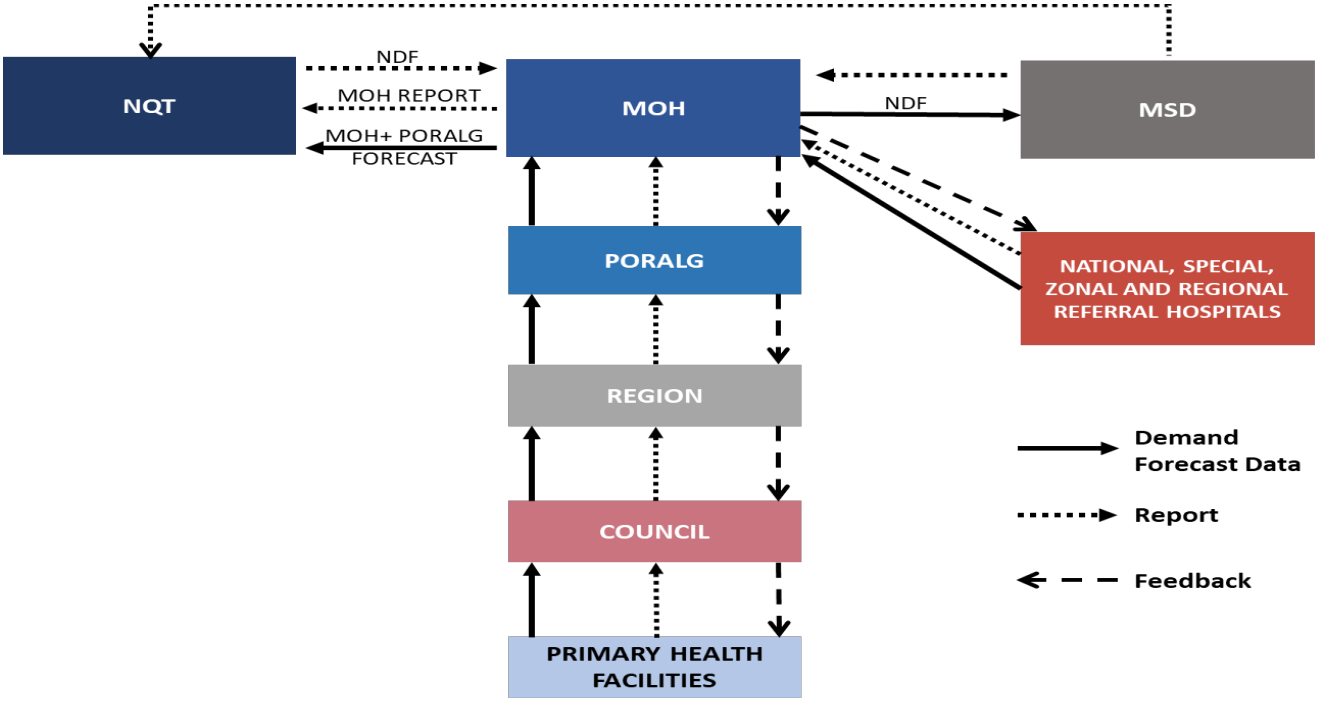
### 6.1.3 Demand Forecast Reports Flow

The submission of the final health facility, council, region, PORALG and MoH demand forecast through the automated BUQ in the eLMIS will then be accompanied by an official letter and report signed by the respective authority that is health facility in charge for health facility, District Executive Commission for Council, Regional Administrative Commission for regional demand forecasts and Permanent Secretary for PORALG and MoH (Refer letter and report template at annex 8.3 below). MSD will also be responsible for submitting the status reports of the supply plan and procurement to NQT quarterly.

### 6.1..4 **Feedback Mechanism**

Different mechanisms can be used to provide feedback on the reviewed demand forecasts. These include direct feedback from council to health facilities, either by indicating remarks in eLMIS automated-BUQ module, or by physically engaging with facilities to incorporate mentoring while also providing feedback on the facility demand forecasts. Secondly, Regional level can provide feedback to the respective council teams while specifying the facility demand forecasts that need to be reviewed. Lastly, the PS-MOH can provide feedback of the reviewed and approved demand forecasts to DCS on MOH-specific hospitals, as well as to the PS-PORALG for PORALG-specific health facilities.

**Figure 3 Flow of demand forecast data, reports and feedback**



# 7 BUQ MONITORING AND EVALUATION FRAMEWORK

The BUQ monitoring and evaluation framework provides a linear relationship of the use of BUQ resources and the desired end of the BUQ process. It comprises five components which are input, process, output, outcome and impact.

i. Input: shows the resources which need to be invested in the BUQ process,

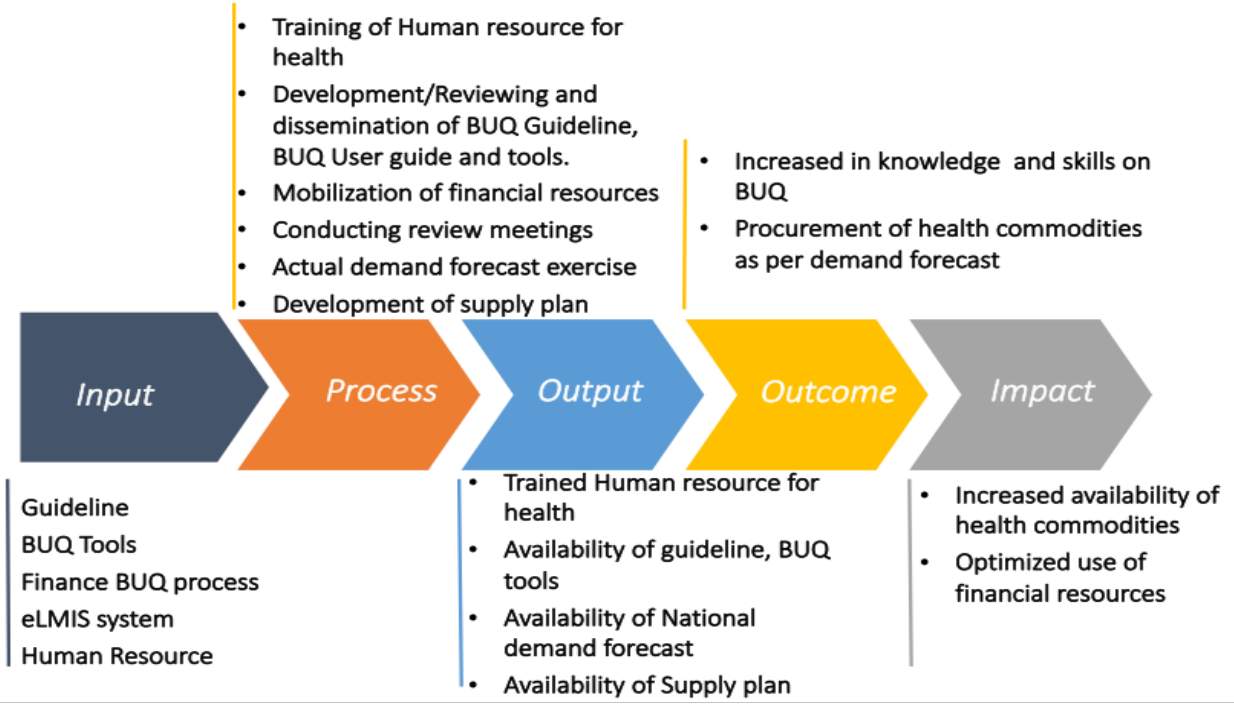
ii. Process: shows the activities which will be carried in the BUQ process,

iii. Output: shows the immediate results after the execution of the BUQ activities

iv. Outcome: shows the short-term or intermediate results achieved after the execution of the BUQ activities, and

v. Impact: shows the results of the BUQ process.

**Figure 4: BUQ monitoring and evaluation framework**



## 7.1 BUQ PERFORMANCE MONITORING

The performance of the BUQ will be measured through the selected indicators which are forecast accuracy and percentage of the health commodities received as per supply plan which will be measured annually and quarterly respectively. The responsibility for performance monitoring should be with the PSU centrally, PORALG centrally, respective RHMTs, CHMTs and Hospital and primary health care Therapeutic Committees.

### i. Forecast accuracy formula

$$\left(1 - \left(\frac{\text{Forecasted consumption in a defined period- Actual Consumption in a defined period}}{\text{Forecasted Consumption in a defined period}}\right)\right) * 100$$

Data source: eLMIS

Data requirements

a. Numerator: eLMIS BUQ Report Forecasted Consumption and BUQ Actual Consumption report

b. Denominator: eLMIS Forecasted Consumption

ii. **Percentage of health commodities received as per supply plan formula**

$$\left(1 - \left(\frac{\text{(Total Number of Items Anticipated in Supply Plan - Number of items received on time and in full per supply plan)}}{\text{Total Number of Items Anticipated in Supply Plan}}\right)\right) * 100$$

Data source: MSD Epicor 10

Data requirementS

a. Numerator: GRN/ Received report from Epicor 10

b. Denominator: - MSD supply plan tool.

**NB:** For further details on calculation of the above indicators, please refer to the public health commodities supply chain indicator reference manual.

### 8.1.2 EVALUATION

The bottom-up demand driven quantification will be subjected to evaluation after the five years of implementation, based on but not limited to the following evaluation criteria;

i. **Effectiveness:** The extent to which Bottom-up quantification has achieve its objective
ii. **Efficiency:** The extent to which the bottom-up quantification has been delivered in an economic and timely way

iii. **Impact:** The long-term effects (negatively and positively) of the bottom-up quantification

iv. **Sustainability:** The extent to which the net benefits of the Bottom-up quantification continue or are likely to continue.

# 8 REFERENCES


1. Essential Health Commodities Quantification Guideline, Tanzania Mainland, July 2018.
2. Comprehensive Council Health Planning (CCHP) Guideline, 5th Edition, December 2020.
3. Guideline for Developing Comprehensive Hospital Operational Plan (CHOP) for Regional Referral Hospitals, Revised Edition, February 2019
4. Medicines and Therapeutics Committee Guideline, Tanzania Mainland, December 2020.
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6. Quantification of health commodities: A guide to forecasting and supply planning for procurement by USAID/DELIVER PROJECT March 2009.
7. Holistic Supply Chain review report, 2017
8. Standard treatment guidelines and National Essential Medicines List June 2021, 6th Edition.
9. National standard for medical laboratories October 2017.
10. Standard Medical Laboratory Equipment Guideline: March, 2018
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# 9 ANNEX

## 9.1 FORECASTING EXAMPLES

### A. EXAMPLE OF CONSUMPTION-BASED FORECASTING METHOD (DETAILED FOR SINGLE PRODUCTS)

**Data available:** Adjusted consumption for July 2022 to June 2023, Paracetamol 500mg (tabs) is 12500 tablets.

Steps to project the forecast using the consumption method

| Guidance | Example |
|---|---|
| Indicate the forecasting assumption that has been selected for the respective product(s). | **Forecast assumptions:**<br>Expect to expand the outpatient medical rooms at the health facility, which will lead to an increase in the number of patients. |
| Determine a quantity factor to assign to the forecasting assumption. Note the quantity factor is based on a consensus among the quantification team (example medicine therapeutic team at the health facility). | Estimated to lead to 10% increase in consumption of Paracetamol 500mg tablets. |
| The historical consumption data, per product, will be multiplied by the quantity factor (2 above) to determine the projected forecast consumption of the next financial year. | i. Historical consumption data, per product= 12500<br>ii. Quantity factor= 10%<br>iii. 12500 x 10% = 125 tablets of increased Paracetamol consumption<br>iv. Total forecasted consumption = 12500 + 125 = 12650 tablets |

## B. EXAMPLE OF CONSUMPTION-BASED FORECASTING METHOD (SUMMARIZED FOR MULTIPLE PRODUCTS)

| Products | Adjusted historical consumption | Forecasted assumption | Total forecasted consumption |
|---|---|---|---|
| Paracetamol 500mg tablets | 12500 tabs | 10% (increase) | 12650 tabs |
| Ceftriaxone 250mg vials | 500 vials | 5% (decrease by controlling AMR at the facility) | 475 vials |

## C. EXAMPLE OF MORBIDITY-BASED FORECASTING METHOD (DETAILED FOR SINGLE PRODUCTS)

**Data available:** Adjusted consumption for July 2022 to June 2023, Paracetamol 500mg (tabs) is 12500 tablets.

Steps to project the forecast using the consumption method

| Guidance | Example |
|---|---|
| **a. Conversion** | |
| List the diseases or health conditions that were identified during the preparation stage, and the number of cases recorded per disease or health condition | ● Disease: Non-severe pneumonia in children<br>● Number of cases: 100<br>● Period: July 2022 to June 2023 |
| Indicate all the products that will be administered in addressing the identified disease or health condition | Products used for pharmacological treatment of non-severe pneumonia in children:<br>● Amoxicillin dispersible tablets<br>● Paracetamol 500mg tablets |
| Determine the quantity of products that are expected to be administered or consumed for the disease or health condition, as indicated in the STG, SMLEG, SMLEL,SMRIEG,disease specific | Quantity of products needed:<br>● Amoxicillin dispersible tablets: 250mg x 2 tablets x 3 times x 5 days = 30 tablets<br>● Paracetamol 500mg tablets: 500mg x 1 tablet x 3 times x 3 |

| Guidance | Example |
|---|---|
| treatment guidelines or testing algorithms. | days = approximately 10 tablets |
| Multiply the products (in 2 above) per listed diseases or health conditions (in 1 above), by the quantity of products to be administered per guideline (in 3 above). The result is the historical consumption that has been converted from number of cases of diseases or health conditions | • Amoxicillin dispersible tablets = 30 tablets x 100 cases = 3000 tablets<br>• Paracetamol 500mg tablets = 10 tablets x 100 cases = 1000 tablets |
| **b. Projection** | |
| Indicate the forecasting assumption that has been selected for the disease or health condition | Forecast assumptions:<br>Expect to expand the outpatient medical rooms at the health facility, which will lead to an increase in the number of patients. |
| Determine a quantity factor to assign to the forecasting assumption. | Estimated to lead to 10% increase in consumption of Paracetamol 500mg tablets. |
| The historical consumption, per product, will be multiplied by the quantity factor (2 above) to determine the projected forecast consumption of the next financial year. | Amoxicillin dispersible tablets<br>i. Historical consumption data= 3000 tablets<br>ii. Quantity factor= 10%<br>iii. 3000 x 10% = 300 tablets of increased consumption<br>iv. Total forecasted consumption = 3000 + 300= 3300 tablets<br>v. Paracetamol 500mg tablets<br>vi. Historical consumption data= 1000 tablets<br>vii. Quantity factor= 10%<br>viii. 1000 x 10% = 100 tablets of increased consumption<br>ix. Total forecasted consumption = 1000 + 100= 1100 tablets |

## D. EXAMPLE OF MORBIDITY-BASED FORECASTING METHOD (SUMMARIZED FOR MULTIPLE PRODUCTS)

| Disease / Condition | Number of cases July 2022 - July 2023 | Products | Quantity of products needed | Forecasted assumption | Total forecasted consumption |
|---|---|---|---|---|---|
| Non-severe pneumonia | 100 | Amoxicillin dispersible tablets | 3000 tabs | 10% (increase) | 3300 tabs |
| | | Paracetamol 500mg tablets | 1000 tabs | 10% (increase) | 1100 tabs |

## E. EXAMPLE OF PROXY-BASED FORECASTING METHOD (DETAILED FOR SINGLE PRODUCTS)

**Data available:** Issues data from the store to the dispensing area for July 2022 to June 2023, Cetrizine 10mg (tabs) is 3000 tablets.

Steps to project the forecast using the proxy-based method

| Guidance | Example |
|---|---|
| Indicate the forecasting assumption that has been selected for the respective product(s). | Forecast assumptions:<br>• All the issued product is consumed<br>• Due to climatic changes, it is estimated that the dry season will be prolonged in the coming year leading to more allergies |
| Determine a quantity factor to assign to the forecasting assumption. Note the quantity factor is based on a consensus among the quantification team | • 100% of the issued Cetrizine is consumed<br>• Estimated to lead to 5% increase in consumption of |

| Guidance | Example |
|---|---|
| (example medicine therapeutic team at the health facility). | Cetrizine 10mg tablets. |
| The proxy consumption data, per product, will be multiplied by the quantity factor (2 above) to determine the projected forecast consumption of the next financial year. | ● Historical issue data, per product= 3000<br>● Quantity factor= 5%<br>● 3000 x 5% = 150 tablets of increased Cetrizine proxy consumption<br>● Total forecasted consumption = 3000 + 150 = 3150 tablets |

## F. EXAMPLE OF PROXY-BASED FORECASTING METHOD (SUMMARIZED FOR MULTIPLE PRODUCTS)

| Products | Proxy consumption | Forecasted assumption | Total forecasted consumption |
|---|---|---|---|
| Cetrizine 10mg tabs | 3000 tabs | 5% (increase) | 3150 tabs |
| ALU 4*4 tabs | 500 strips | 10% (decrease through improved community interventions) | 450 vials |
| Ringer Lactate solution | 4000 bottles | No change | 4000 bottles |

## G. EXAMPLE OF SERVICE LEVEL PROJECTION OF REQUIREMENTS BASED FORECASTING METHOD (DETAILED FOR SINGLE PRODUCTS)

**Data available:** Consumption data from health facility A for July 2022 to June 2023, Ibuprofen 200mg (tabs) is 11000 tablets.

Steps to project the forecast using the consumption method

| Guidance | Example |
|---|---|
| Indicate the forecasting assumption that has been selected for the respective product(s). | Forecast assumptions:<br>Health Facility A has similar characteristics in terms of type of facility, disease burden, catchment area, and types of services provided as health facility B. |
| Determine a quantity factor to assign to the forecasting assumption. Note the quantity factor is based on a consensus among the quantification team (example medicine therapeutic team at the health facility). | There will be no change in the amount of Ibuprofen tablets consumed in the coming year. |
| The service level data, per product, will be multiplied by the quantity factor (2 above) to determine the projected forecast consumption of the next financial year. | ● Service level data, per product= 11000<br>● Quantity factor= 0%<br>● 11000 x 0% = 0 tablets of increased Ibuprofen tablets consumption<br>● Total forecasted consumption = 11000 + 0 = 11000 tablets |

## H. EXAMPLE OF SERVICE LEVEL PROJECTION OF REQUIREMENTS BASED FORECASTING METHOD (SUMMARIZED FOR MULTIPLE PRODUCTS)

| Products | Service level Projection | Forecasted assumption | Total forecasted consumption |
|---|---|---|---|
| Ibuprofen 200mg tabs | 11000 tabs | No change | 11000 tabs |
| Omeprazole 20mg tabs | 7200 tabs | 15% increase | 8280 tabs |
| Hydrocortisone 100mg inj | 3000 vials | 5% increase | 3150 vials |

## 5.1 FOMU YA TAARIFA YA MAANDALIZI YA MAOTEO YA BIDHAA ZA AFYA KUTOKA HALMASHAURI KWENDA MKOANI

### TAARIFA YA MAANDALIZI YA MAOTEO YA BIDHAA ZA AFYA

| JINA LA HALMASHAURI | MKOA | TAREHE |
|---|---|---|
| | | |

1. **UTANGULIZI :**

Toa maelezo mafupi ya shughuli ya maoteo ilivyofanywa. Gusia kuhusu;

1.1 Tarehe ya kuanza maoteo kuanzia tarehe ……..... hadi tarehe .....

1.2 Shughuli zilizotangulia kabla ya zoezi halisi la maoteo mfano vikao vya maandalizi nk.

.....
.....
.....
.....

2. **WATAALAMU WALIOSHIRIKI KATIKA MAOTEO;**

**Jedwali Na.1: Walioshiriki maandalizi ya Maoteo ya Bidhaa za Afya Ngazi ya Wilaya**

| Na. | Jina Kamili | Taaluma | Cheo |
|---|---|---|---|
| 1. | | | |
| 2. | | | |
| 3. | | | |
| 4. | | | |
| 5. | | | |

3. Eleza kama shughuli hii iliwekwa kwenye mpango wenu wa utekelezaji wa mwaka CCHP ……..……

4. Kama kuna support mliyopata ya wadau katika kipindi hiki itaje na ueleze ilivyosaidia mchakato ……

**5.** Jambo lolote lingine kwa ufupi …….

**6. MAANDALIZI YA MAOTEO**

Eneo hili fafanua kuhusu mambo yafuatayo

i. Hatua za maandalizi ya maoteo (hatua kwa hatua kwa ufupi)

ii. Vyanzo vya taarifa vilivyotumika katika kufanya maoteo. Taja aina ya chanzo cha taarifa na aina ya taarifa uliyotumia kutoka katika vyanzo hivyo kwa mchanganuo ufuatao:

**Jedwali na. 2: VYANZO VYA TAARIFA VILIVYOTUMIKA**

| CHANZO CHA TAARIFA | AINA YA TAARIFA |
|---|---|
| Mfano: eLMIS | Taarifa ya matumizi ya bidhaa za afya kwa mwaka 2022-2023 |
| | |
| | |
| | |

iii. Assumptions (nadharia) mlizoweka wakati wa maandalizi ya maoteo (MUHIMU SANA HILI)

iv. Toa summary ya vituo vilivyofanya maoteo mwaka huu kwa mchanganuo ufutao

**Jedwali Na. 3: Idadi ya vituo vilivyowasilisha maoteo kwa mwaka wa Fedha 2023/2024**

| AINA YA VITUO | IDADI YA VITUO VINAVYOFANYA KAZI SASA | IDADI AMBAVYO VINATARAJIWA KUANZA KAZI | JUMLA |
|---|---|---|---|
| Zahanati | | | |
| Vituo Vya Afya | | | |
| Hospitali | | | |
| Vingine (Mf. Duka la Dawa N.K) | | | |
| JUMLA | | | |

## MUHTASARI YA THAMANI YA BIDHAA ZILIZOOTEWA KAMA WILAYA NA KWA MCHANGANUO UFUATAO

**Jedwali Na. 4: Thamani ya Bidhaa za Afya zilizootewa kwa mwaka wa Fedha ……**

| NA | NGAZI YA KITUO CHA HUDUMA | THAMANI YA BIDHAA KWA KILA KUNDI LA BIDHAA | | | | | |
|---|---|---|---|---|---|---|---|
| | | Dawa | Vifaa na Vifaa Tiba | Vitendanishi vya Maabara | Bidhaa na vifaa vya meno | Bidhaa za Radiolojia | JUMLA |
| 1. | Hospitali | | | | | | |
| 2. | Vituo vya Afya | | | | | | |
| 3. | Zahanati | | | | | | |
| 4. | Vituo Vinginevyo (Mfano ; Duka la Dawa N.K) | | | | | | |
| | JUMLA | | | | | | |

7. **CHANGAMOTO ZA MAANDALIZI YA MAOTEO**

i. .....
ii. .....
iii. .....

8. **MAPENDEKEZO**

i. .....
ii. .....
iii. .....

## 5.2 VIAMBATANISHO

- Barua kutoka kwa Mkurugenzi Wa Halmashauri ya kuidhinisha maoteo.
- Orodha ya Vituo vilivyofanya maoteo na thamani ya maoteo hayo

| NA | JINA LA KITUO CHA KUTOLEA HUDUMA | THAMANI YA BIDHAA ZILIZOOTEWAKA (TZS) | BAJETI YA BIDHAA ZA AFYA KWA MUJIBU WA MPANGO WA KITUO (TZS) |
|---|---|---|---|
| | | | |
| | | | |

## MFANO WA BARUA YA KUWASILISHA MAOTEO

**YAH: KUWASILISHA MAOTEO YA BIDHAA ZA AFYA KWA MWAKA ……**

Tafadhari rejea kichwa cha habari hapo juu,

Kwa mujibu wa Sheria ya Manunuzi, vituo vya kutolea huduma za afya vinatakiwa kuandaa maoteo yake ya bidhaa za afya na kuyawasilisha Bohari ya Dawa (MSD) kila mwaka kwa kufuata utaratibu maalumu uliowekwa na Muongozo wa maandalizi wa Maoteo hayo.

Halmashauri ya ……. imefanya maoteo ya bidhaa za afya zitakazohitajika katika mwaka wa fedha …… katika vituo vyake ..... (Taja Idadi). Mchanganuo wa vituo hivi umeainishwa katika Jedwali Na. 3, 4 na 5 (taarifa iliyoambatishwa). Maoteo haya yana jumla ya Thamani ya TZS …….…… Hata hivyo, kwa kuzingatia mwenendo wa bajeti yetu ya mwaka wa fedha ……..., Halmashauri itaweza kugharimikia bidhaa zenye thamani ya TZS ……. Kutoka katika vyanzo vya papo kwa papo, NHIF, ICHF, DRF, n.k na hivyo kuwa na upungufu wa fedha kiasi cha TZS ……

Kwa barua hii, naomba kuwasilisha taarifa fupi ya maandalizi ya maoteo hayo baada ya kutuma maoteo ya vituo vyote kwa njia ya kielektroniki ya eLMIS.

Natanguliza shukurani zangu

……..
Mkurugenzi

## 5.3 TAARIFA YA MAANDALIZI YA MAOTEO YA BIDHAA ZA AFYA KUTOKA MKOANI KWENDA TAMISEMI

### TAARIFA YA MAANDALIZI YA MAOTEO YA BIDHAA ZA AFYA

JINA LA MKOA ……..... TAREHE ……

### 1. UTANGULIZI:

Toa maelezo mafupi ya shughuli ya maoteo ilivyofanywa:

1.1 Tarehe ya kuanza maoteo ..... hadi .....

1.2 Shughuli zilizotangulia kabla ya zoezi halisi la maoteo mfano vikao vya maandalizi nk.

.....
.....
.....
.....
.....

### 2. Wataalamu walioshiriki katika maoteo:

**Jedwali Na.1: Walioshiriki maandalizi ya maoteo ya bidhaa za Afya ngazi ya Mkoa**

| Na | Jina Kamili | Taaluma | Cheo |
|---|---|---|---|
| 1. | | | |
| 2. | | | |
| 3. | | | |
| 4. | | | |

3. Eleza kama shughuli hii iliwekwa kwenye mpango wenu wa utekelezaji wa mwaka …….

4. Kama kuna support mliyopata ya wadau katika kipindi hiki itaje na ueleze ilivyosaidia mchakato ……..
……..
……..……
……..……

5. Jambo lolote linguine kwa ufupi ……..….

6. **MAANDALIZI YA MAOTEO**

Eneo hili fafanua kuhusu mambo yafuatayo

i. Hatua za maandalizi ya maoteo (hatua kwa hatua kwa ufupi)

ii. Vyanzo vya taarifa vilivyotumika katika kufanya maoteo. Taja aina ya chanzo cha taarifa na aina ya taarifa uliyotumia kutoka katika vyanzo hivyo kwa mchanganuo ufuatao:

**Jedwali na. 2: VYANZO VYA TAARIFA VILIVYOTUMIKA**

| CHANZO CHA TAARIFA | AINA YA TAARIFA |
|---|---|
| Mfano: eLMIS | Taarifa ya matumizi ya bidhaa za afya kwa mwaka |
| | |
| | |
| | |

iii. Assumptions (nadharia) mlizoweka wakati wa maandalizi ya maoteo

iv. Toa summary ya vituo vilivyofanya maoteo mwaka huu kwa mchanganuo ufutao:

**Jedwali Na. 3: Idadi ya vituo vilivyowasilisha maoteo kwa mwaka wa fedha ……...**

| AINA YA VITUO | IDADI YA VITUO VINAVYOFANYA KAZI SASA | IDADI AMBAVYO VINATARAJIWA KUANZA KAZI | JUMLA |
|---|---|---|---|
| Zahanati | | | |
| Vituo Vya Afya | | | |
| Hospitali | | | |
| Vituo Vinginevyo (Mfano ; Duka La Dawa N.K) | | | |
| JUMLA | | | |

**MUHTASARI WA THAMANI YA BIDHAA ZILIZOOTEWA KAMA MKOA KWA MCHANGANUO UFUATAO:**

**Jedwali Na. 4: Thamani ya Bidhaa za Afya zilizootewa kwa mwaka wa Fedha ……**

| NA | NGAZI YA KITUO CHA HUDUMA | THAMANI YA BIDHAA KWA KILA KUNDI LA BIDHAA | | | | | |
|---|---|---|---|---|---|---|---|
| | | Dawa | Vifaa na Vifaa Tiba | Vitendanishi vya Maabara | Bidhaa na vifaa vya meno | Bidhaa za Radiolojia | JUMLA |
| 1. | Hospitali | | | | | | |
| 2. | Vituo vya Afya | | | | | | |
| 3. | Zahanati | | | | | | |
| 4. | Nyinginezo (Fafanua) | | | | | | |
| | JUMLA | | | | | | |

## 7. CHANGAMOTO ZA MAANDALIZI YA MAOTEO

i. .....
ii. .....
iii. .....

## 8. MAPENDEKEZO

i. .....
ii. .....

## 6. VIAMBATISHO

- Barua ya Katibu Tawala wa Mkoa kuidhinisha maoteo ya Mkoa.
- Orodha ya Halmashauri zilizofanya Maoteo na Thamani ya Maoteo hayo

| NA | JINA LA HALMASHAURI | THAMANI YA BIDHAA ZILIZOOTEWAKA (TZS) | BAJETI YA BIDHAA ZA AFYA KWA MUJIBU WA MPANGO WA KITUO (TZS) |
|---|---|---|---|
| | | | |
| | | | |
| | | | |

## 8.5 VIGEZO VYA KUPITIA TAARIFA ZA MAOTEO

| NA. | KIGEZO KILICHOANGALIWA |
|---|---|
| 1 | Ukamilifu wa takwimu za maoteo zilizowasilishwa (Data completeness) |
| 2 | Usahihi wa takwimu zilizowasilishwa (Data Accuracy) |
| 3 | Uwepo wa dawa muhimu (Check Essential Medicines (30/290); Reporting and Quantities |
| 4 | Mahitaji ya bidhaa muhimu za kiuchunguzi wa afya (Check Essential Diagnostics (Standardized Equipment/Reagents) |
| 5 | Vyanzo vya mapato (NHIF, UF, CHIF, RK, BF, COs, Others) and Funding Gap |
| 6 | Matumizi sahihi ya nyenzo mpya ya maoteo (Check Compliance of Council Tool (PHC and Council Hospital) |
| 7 | Mahitaji maalum kwa hospitali maalum au kituo kingine chochote (Check Special items for Specialized Hospitals and any other facility) |
| 8 | Hakiki na tambua takwimu ambazo haziendani na hali halisi (“outliers”) |